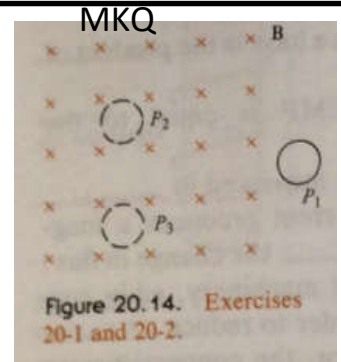


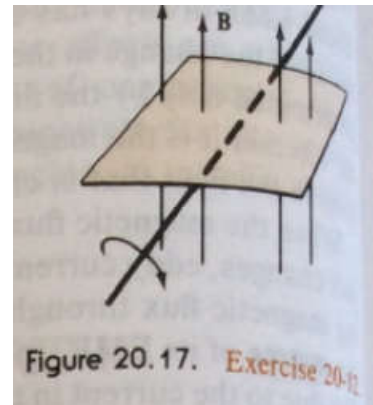
20.1 In Fig. 20.14 the magnetic field is uniform in the region shown and is into the page. A wire loop is in the plane of the page. State whether there is and induced current when the loop is re-moved from (a) P1 to P2 (b) P2 to P3 (C) P3 to P1. Give reasons from your answers.



20.2 The wire loop in Fig. 20.14 is moved in the plane of the page from P1 to P2. Find the direction of the induced current, if any.

20.5 A wire loop of radius 0.1m has a 2Ω resistance. A uniform magnetic field perpendicular to the plane of the loop increases from zero to 2T in 0.01s. Find the average induced current.

20.6 A circular wire loop has a resistance of 10Ω and of a radius of 0.1m. Its normal is parallel to a uniform magnetic field of magnitude 3T. If the loop is rotated so that in 0.3s its normal is perpendicular to the field, find (a) the average induced emf (b) the average induced current.



20.12 A square wire loop of resistance 10Ω with side 0.2m rotates 100 times per second about a horizontal axis (Fig. 20.17). The magnetic field is vertical and has a magnitude of 0.5T. Find the amplitude of the induced current.

20.14 The tube of an X-ray machine requires an ac voltage of 50000V. If this is obtained from 120V line, what is the ratio N_2/N_1 of the number of turns in the secondary and primary coils of the transformer?

20.16 A transformer has 100 turns in one coil and 500 in the other. It is connected to a 120V ac power line. What is the voltage across the secondary if (a) the 100 turn coil is connected to power line (b) the 500 turn coil is connected to the power line?

20.33 A bar magnet is dropped through a wire loop (Fig. 20.20). Find the direction of the induced current when the magnet is (a) above the loop (b) halfway through the loop (C) below the loop. Draw a rough graph of the induced current versus time, assuming the velocity of the magnet is constant.

